

Absolutely. Based on your progress and your goal (**becoming an AI freelancer who builds AI agents and automation systems**), here's the complete roadmap I would mentor you through.

This is **not** a generic roadmap from YouTube. It's structured to take you from your current level (Node.js + React + AI basics) to building production AI applications.

Phase 1 — AI Fundamentals (Completed)

Duration: 1 week

Module 1: LLM Fundamentals

-  What is an LLM?
-  Tokens
-  Context Window
-  Temperature
-  How LLMs generate text

Module 2: Prompt Engineering

-  Roles
-  Structured Output
-  Prompt Improvement
-  System vs User prompts
-  Prompt design principles

Module 3: OpenAI API

-  API Keys
-  Requests & Responses
-  `response.output_text`
-  Environment Variables
-  Cost & Tokens

Module 4: Memory

-  Stateless LLMs
-  Conversation History
-  `messages` array
-  Database-backed memory

Module 5: RAG Fundamentals

-  Chunking

-  Embeddings
 -  Vector Databases
 -  Similarity Search
 -  Top-K Retrieval
 -  RAG Architecture
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Phase 2 — Build Production AI Apps (Current)

Duration: 4–5 weeks

Project 1 — AI Chatbot Backend

- Express setup
 - Folder structure
 - Routes
 - Controllers
 - Services
 - Utils
 - Chat endpoint
 - Conversation history
 - MySQL/PostgreSQL integration
 - Multi-user support
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Project 2 — PDF RAG Chatbot

Week 1

- Upload PDF
- Extract text
- Parse PDF
- Handle errors

Week 2

- Chunking
- Chunk overlap
- Metadata
- Token counting

Week 3

- Generate embeddings

- Embedding models
- Vector database
- Store embeddings

Week 4

- Similarity search
 - Top-K Retrieval
 - Build prompt
 - Ask questions
 - Return answers
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Project 3 — Multi-Document Knowledge Base

Features:

- Upload multiple PDFs
 - Multiple users
 - Search across documents
 - Source citations
 - Page references
 - Conversation history
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Project 4 — Streaming Chatbot

Learn:

- Streaming responses
 - Typing effect
 - Server-Sent Events (SSE)
 - Real-time updates
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Project 5 — Production Improvements

- Authentication
- Rate limiting
- Logging
- Error handling
- Validation
- Docker basics
- Deployment



Phase 3 — AI Engineering

Duration: 3–4 weeks

Function Calling

- What is tool calling?
 - When should AI call a tool?
 - JSON schema
 - Calling backend functions
 - Multi-tool workflows
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LangChain

Because you'll already know the fundamentals, LangChain will feel like a helper instead of magic.

Learn:

- PromptTemplate
- OutputParser
- DocumentLoader
- TextSplitter
- Embeddings
- VectorStore
- Retriever
- Chains
- Memory
- Agents

Build:

- LangChain RAG
 - Website chatbot
 - SQL chatbot
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LangGraph

Learn:

- Graphs
- Nodes

- Edges
- State
- Conditional routing
- Loops
- Interrupts

Build:

- Research Agent
 - Planning Agent
 - Multi-step workflows
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MCP (Model Context Protocol)

Learn:

- What MCP is
- MCP Clients
- MCP Servers
- Tool discovery
- Resource sharing
- Security
- Local vs Remote MCP

Build:

- AI agent using MCP tools
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Multi-Agent Systems

Build:

- Manager Agent
 - Research Agent
 - Coding Agent
 - Review Agent
 - Report Generator
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Phase 4 — AI Automation (n8n)

Duration: 2–3 weeks

n8n Fundamentals

- Nodes
 - Workflows
 - Triggers
 - Webhooks
 - Credentials
 - Expressions
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AI Nodes

- OpenAI Node
 - AI Agent Node
 - Memory Node
 - Tool Node
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Integrations

- Gmail
 - Google Sheets
 - Calendar
 - Slack
 - Discord
 - Notion
 - Airtable
 - CRM
 - WhatsApp
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Projects

- Lead Generation Agent
 - Resume Screening
 - Customer Support
 - Email Assistant
 - Invoice Processing
 - Social Media Automation
 - Meeting Notes Agent
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Learn:

- Hybrid Search
- BM25
- Metadata Filtering
- Parent-Child Retrieval
- Multi-Query Retrieval
- Query Expansion
- Reranking
- Context Compression
- Caching
- Cost Optimization

Build:

- Enterprise Knowledge Assistant
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Phase 6 — AI Deployment & Production

Learn:

- Docker
- Environment variables
- Reverse proxy basics
- Deployment
- Monitoring
- Logging
- Token usage tracking
- Cost optimization
- Security best practices

Deploy:

- RAG chatbot
 - AI agent
 - n8n workflows
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Phase 7 — Freelancing & Agency

Learn:

- Portfolio creation
- LinkedIn optimization
- GitHub portfolio

- Upwork profile
 - Fiverr gigs
 - Proposal writing
 - Discovery calls
 - Pricing
 - Client communication
 - Requirement gathering
 - Project estimation
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Phase 8 — Capstone Portfolio Projects

By the end, you'll have projects like:

AI Applications

- ☒ AI Chatbot
- ☒ PDF RAG Chatbot
- ☒ Multi-PDF Knowledge Assistant
- ☒ SQL AI Assistant
- ☒ Website Chatbot

AI Agents

- ☒ Research Agent
- ☒ Resume Screening Agent
- ☒ Email Agent
- ☒ Meeting Assistant
- ☒ Multi-Agent Workflow

Automation

- ☒ Gmail AI Automation
 - ☒ WhatsApp Automation
 - ☒ CRM Automation
 - ☒ Lead Generation
 - ☒ Invoice Processing
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Optional (After You Start Earning)

These are excellent skills, but I recommend learning them after you're already building projects and getting freelance work:

- Fine-tuning LLMs

- Local LLMs with Ollama
 - LlamaIndex
 - AI evaluation and benchmarking
 - AI observability
 - Guardrails and safety
 - Voice AI
 - AI image generation
 - AI coding assistants
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Final Outcome

By the end of this roadmap, you'll be able to:

-  Build AI chatbots from scratch.
-  Build production RAG systems.
-  Develop AI agents with tools and memory.
-  Create AI automations using n8n.
-  Deploy production-ready AI applications.
-  Explain your architecture in interviews.
-  Confidently bid on freelance AI projects.

My Mentoring Approach

Most courses teach you **frameworks first**.

I'm deliberately teaching you **engineering first**.

That means when a new framework or model is released, you'll be able to understand it quickly because you already know the underlying architecture. That's a much more durable skill set for a career in AI engineering.